

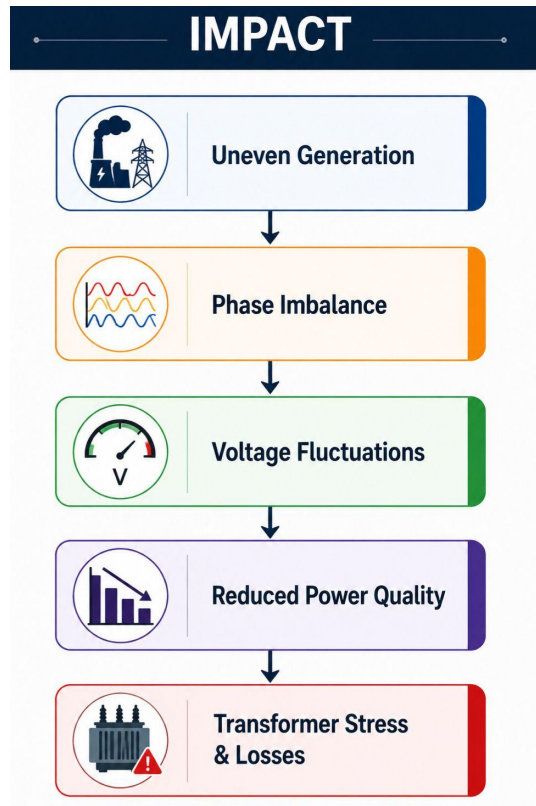
TRIPHASE SMARTGRID

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- Renewable energy adoption is rapidly increasing across Kerala..
 - Rooftop Solar PV systems are widely installed in residential buildings.
 - Most rooftop solar systems are connected to a single phase (R, Y, or B).
 - Uneven solar power injection creates phase imbalance in distribution feeders..
 - Phase imbalance leads to voltage fluctuations and increased neutral current..
 - It also causes transformer stress, power losses, and poor power quality.
 - Existing balancing methods are mostly manual and inefficient.
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Challenges in Existing Distribution Feeders

- Uneven renewable energy generation leads to phase imbalance.
- Certain phases experience excessive loading and overload conditions.
- Available capacity in other phases remains underutilized.
- Voltage fluctuations affect system stability and performance.
- Poor load distribution reduces overall power quality.
- Continuous phase imbalance increases transformer stress and losses.
- Manual load management is inefficient and time-consuming.



- A smart phase balancing system is developed using ESP32, voltage sensors, current sensors, and relay modules.
- The system continuously monitors the voltage and current of all three phases in real time.
- Phase conditions are analyzed to determine which phase is the healthiest and least loaded.
- When a phase experiences low voltage or high loading, the system identifies a better available phase.
- Relay modules automatically shift selected loads to the healthier phase.
- This helps distribute loads more evenly across the three phases.
- The system maintains stable voltage conditions and reduces the risk of phase overloading.
- Real-time system parameters and switching decisions are displayed on the OLED display.

KEY FEATURES



No Cloud
Dependency



No Internet
Required



Local Decision
Making



Real-Time
Load Shifting



Improved
Phase Balance



Cost-Effective
Solution

Component	Function
ESP32 Development Board	Central controller for data acquisition, AI-based decision making, and relay control
Voltage Sensors	Measure the voltage levels of R, Y, and B phases
ACS712 Current Sensors	Monitor load current in each phase and detect overload conditions
4-Channel Relay Modules	Perform automatic phase switching and load shifting between phases
Potentiometers	Simulate varying renewable energy generation (different phase voltages)
12V LED Bulbs	Represent household/consumer electrical loads
RPC Module	Enables wireless communication for remote monitoring, remote control, or communication between multiple smart grid nodes/controllers.
OLED Display (0.96")	Display phase voltages, currents, and load-shifting status in real time
12V DC Power Supply	Provides power to the entire prototype
Jumper Wires & Breadboard	Establish electrical connections between components

Phase Health Evaluation

Purpose

The ESP32 continuously evaluates the condition of each phase using real-time voltage and current measurements.

INPUT PARAMETERS

PARAMETER	DESCRIPTION
 Voltage Level	Indicates phase availability and strength
 Load Current	Indicates the loading condition of the phase

HEALTH SCORE CALCULATION

The controller assigns a health score to each phase using:

$$\text{Health Score} = (0.7 \times \text{Voltage Contribution}) - (0.3 \times \text{Current Contribution})$$



VOLTAGE CONTRIBUTION
Higher voltage level increases the health score.



CURRENT CONTRIBUTION
Higher load current decreases the health score.

Decision Logic

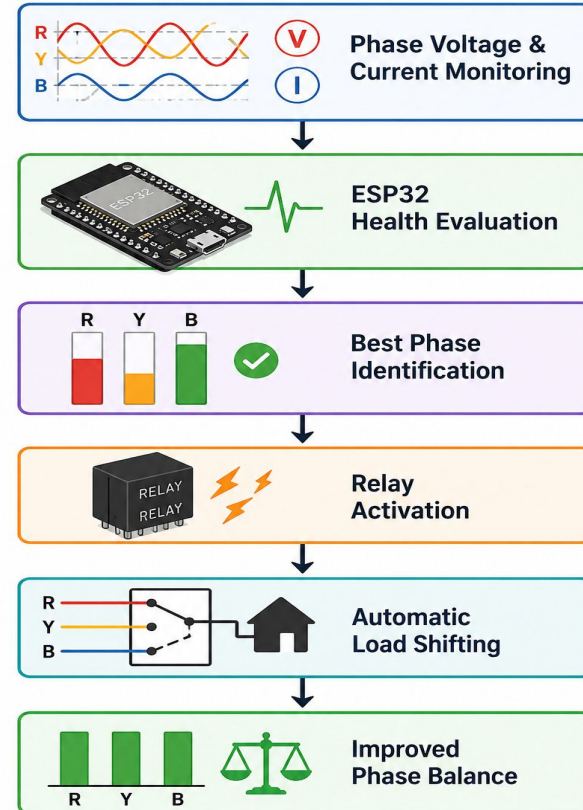
- Higher voltage increases the phase score.
- Higher current decreases the phase score.
- The phase with the highest score is selected for load allocation.

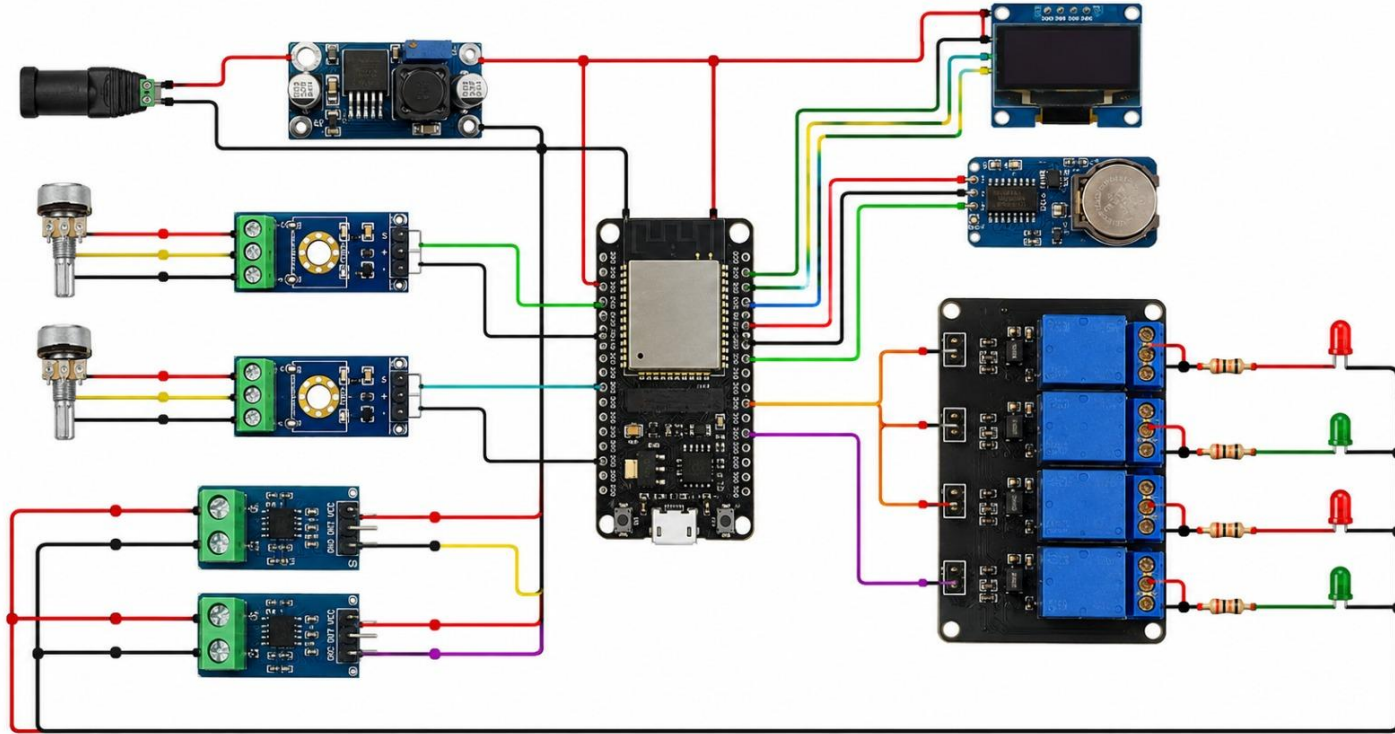
Each load is connected to two alternative phases through relays. The ESP32 continuously evaluates phase health and shifts the load to the healthiest available phase.

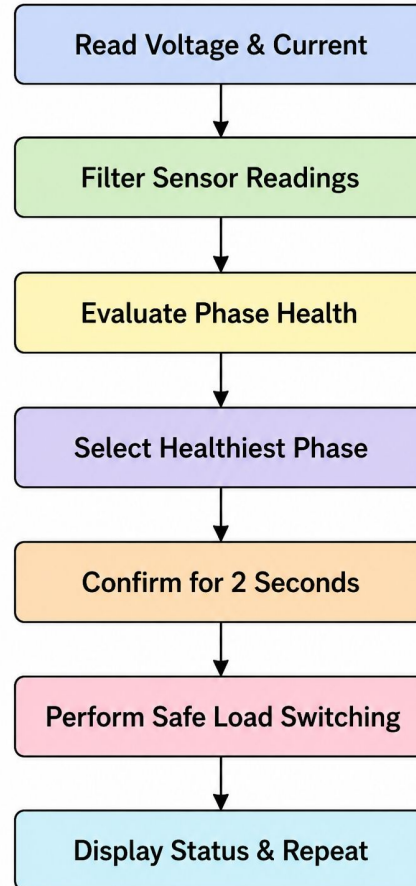
Example load configuration

Load	Primary Phase	Alternate Phase
Bulb 1	R	B
Bulb 2	Y	R
Bulb 3	B	Y

Working Principle







Initial Condition

Parameter	R Phase	B Phase
Voltage	11.8 V	10.6 V
Current	1.2 A	2.8 A
Health Score	High	Low

System decision

- ESP32 reads voltage and current from both phases.
- Health score of **R Phase** is higher.
- Both LED loads remain connected to **R Phase**.
- OLED displays "**System Stable**"

After Voltage Change

Parameter	R Phase	B Phase
Voltage	9.9 V	11.9 V
Current	1.5 A	1.1 A
Health Score	Low	High

System decision

- The ESP32 detects that the **B Phase** has become healthier.
- The condition is verified for **2 seconds** to prevent unnecessary switching.
- The controller safely disconnects the loads from the current phase.
- Both LED loads are automatically connected to the **B Phase**.
- The OLED displays "**Shift Successful: R → B**".

Voltage Threshold Detection

-  Detects weak phases.
-  Prevents load allocation to unstable phases.
-  Ensures reliable operation.

Overload Detection

-  Detects excessive feeder loading.
-  Prevents additional load assignment.
-  Protects system components.

Automatic Isolation

-  Rejects unhealthy phases.
-  Maintains system stability.
-  Improves reliability.

Features

- Simulated Three-Phase Renewable Energy System
- Real-Time Voltage & Current Monitoring
- ESP32-Based Intelligent Decision Making
- Automatic Relay-Based Load Shifting
- Live System Status on OLED Display

Loads Used

6 × 12V LED Bulbs

Represent residential consumer loads for demonstrating phase balancing and load transfer.

ADVANTAGES

	Real-Time Monitoring
	Automatic Phase Selection
	Intelligent Load Shifting
	Improved Load Distribution
	Reduced Overloading
	Low-Cost Implementation

APPLICATIONS

	Smart Grids
	Renewable Energy Systems
	Solar Power Integration
	Rural Electrification
	Industrial Power Management
	Microgrids

TRIPHASE SMARTGRID successfully demonstrates real-time phase monitoring and automatic load shifting using ESP32, sensors, and relays. By selecting the healthiest phase based on voltage and current conditions, the system improves load distribution, enhances reliability, and supports efficient smart grid operation.

Project Achievements

- Real-time monitoring of three-phase conditions
 - Automatic identification of the healthiest phase
 - Relay-based load shifting demonstration
 - Improved load distribution across phases
 - Low-cost and scalable smart grid prototype
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THANK YOU
